Mechanics Of Machinery

Can crusher with a quick-return mechanism

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I. Introduction.

A can crusher with a quick-return mechanism is a mechanical device designed to compress aluminum cans efficiently. It uses a crank and lever system to produce a fast forward stroke for crushing the can and a slower return stroke to reset the ram. This design increases speed and ease of operation. The crusher reduces the volume of cans for recycling, saves storage space, and is commonly used in homes, workshops, and recycling centers.

In this project, we designed and studied a Quick Return Mechanism using SolidWorks for 3D modeling and MATLAB for motion analysis. The goal was to understand how the mechanism works, analyze its movement, and show the quick return effect clearly through simulation. This report explains the design process, how the mechanism moves, and the results of our analysis.

ii. Concept & Working principle:-

The can crusher operates using a **quick return mechanism**, which converts rotary motion of the crank into reciprocating motion of the crusher plate (ram). The mechanism is designed so that the **forward stroke (crushing stroke)** takes more time than the **return stroke**, allowing effective crushing while reducing the total cycle time.

When the crank rotates, motion is transmitted through the connecting links and slotted lever, causing the ram to move linearly along the guide inside the frame. During the forward stroke, the ram applies a compressive force on the aluminum can placed in the crushing chamber. The longer duration of this stroke provides sufficient time and force to deform the can.

The quick return action of the mechanism is represented by the **Quick Return Ratio (QRR)**:

$$\text{QRR} = \frac{\text{Time of forward stroke}}{\text{Time of return stroke}} = \frac{\theta_f}{\theta_r}$$

where θ_f and θ_r are the crank angles corresponding to the forward and return strokes respectively. For a quick return mechanism, QRR > 1, indicating that the return stroke occurs in a shorter time.

After the can is crushed, continued rotation of the crank causes the ram to return rapidly to its initial position, preparing the system for the next cycle.

The ability of the mechanism to generate high crushing force is explained by the **mechanical advantage**:

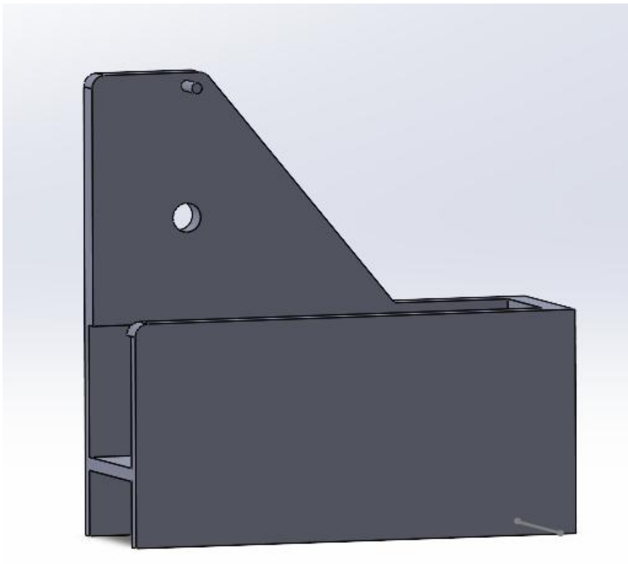
$$\text{Mechanical Advantage (MA)} = \frac{F_{\text{output}}}{F_{\text{input}}}$$

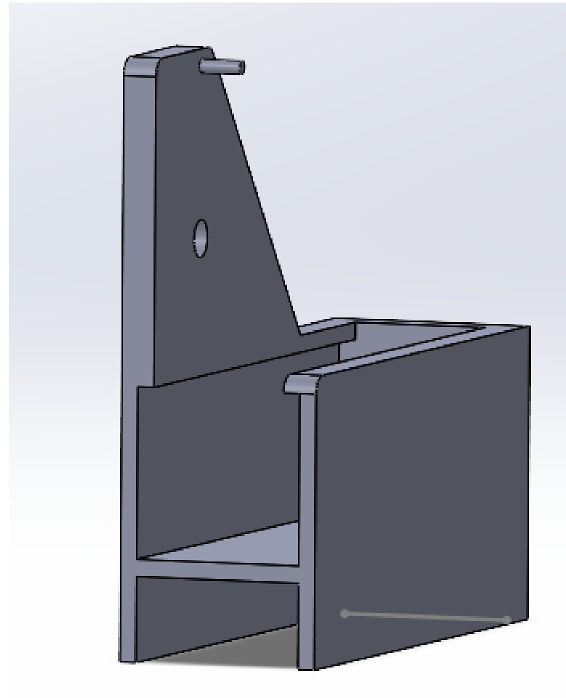
Due to the linkage geometry, the mechanical advantage is higher during the crushing stroke, enabling efficient crushing with relatively low input force.

iii. Mechanical design:-

1) **Frame (Fixed Link):** The frame acts as the stationary member of the mechanism. It supports all moving parts and absorbs the reaction forces developed during the crushing operation.
2) **Crank (Driving Link):** The crank is the rotating link connected to the input shaft or motor. It provides the initial rotary motion required to drive the mechanism.
3) **Connecting Rod (Coupler Link):** The connecting rod transmits motion from the crank to the slotted lever. It converts the rotary motion of the crank into oscillatory motion of the slotted lever.
4) **Slotted Lever:** The slotted lever allows sliding motion of the connecting rod within the slot. This sliding action is responsible for producing unequal time intervals for the forward and return strokes, resulting in the quick return mechanism.
5) **Ram / Slider (Crushing Ram)**: The ram is the reciprocating member that moves linearly along the guide provided by the frame. It applies compressive force on the can during the forward stroke and returns rapidly during the return stroke.

iv. SolidWorks design:-





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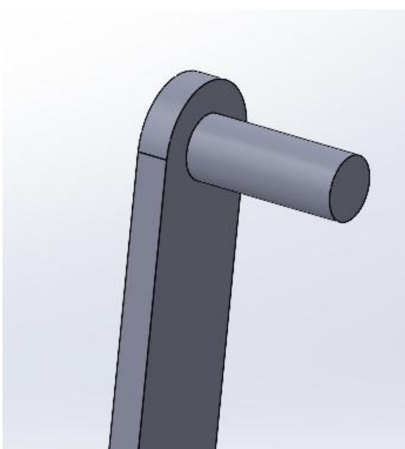

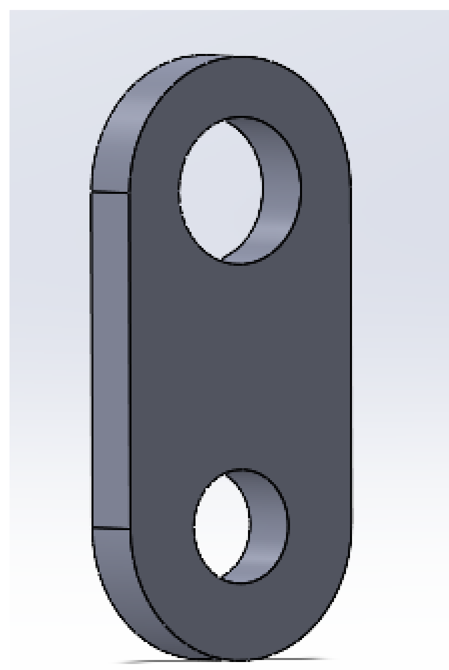

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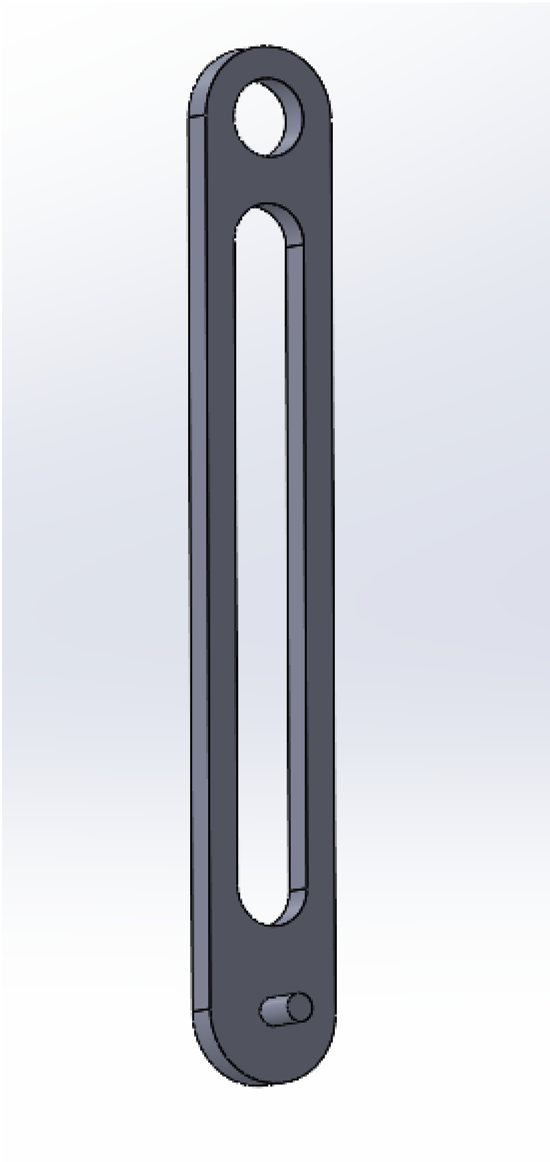

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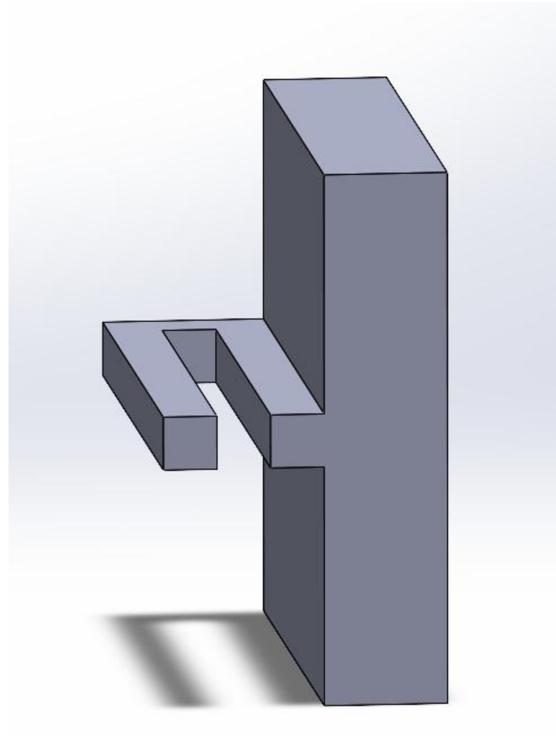

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v. Handwritten analysis.

Position Analysis:

$\overrightarrow{OA} = x_A + i y_A$ $\qquad x_q = 0$

$\rightarrow x_A = L_2 \cos\theta_2$ $\qquad y_q = L_3$

$\rightarrow y_A = L_2 \sin\theta_2$

Loop OAQ:

$\overrightarrow{OA} = \overrightarrow{OQ} + \overrightarrow{QA}$

$x_A + i y_A = x_q + i y_q + x e^{i\theta_3}$

$x e^{i\theta_3} = [x_A - x_q] + i[y_A - y_q]$

$\longrightarrow x = \sqrt{(x_A - x_q)^2 + (y_A - y_q)^2}$

$\longrightarrow \cos\theta_3 = \frac{x_A - x_q}{x}$, $\sin\theta_3 = \frac{y_A - y_q}{x}$

Loop OBQ:

$\overrightarrow{OB} = \overrightarrow{OQ} + \overrightarrow{QB}$

$x_B + i y_B = x_q + i y_q + L_4 e^{i\theta_3}$

equating real parts:

$\longrightarrow x_B = x_q + L_4 \cos\theta_3$

equating imaginary parts:

$\rightarrow y_B = y_q + L_4 \sin\theta_3$

Loop OCB:

$x_c + i y_c = x_B + i y_B + L_5 e^{i\theta_4}$

equating imaginary parts:

$y_c = y_B + L_5 \sin\theta_4$

$\longrightarrow \sin\theta_4 = \frac{y_c - y_B}{L_5}$

equating real parts:

$\rightarrow x_c = x_B + L_5 \cos\theta_4$

Loop ODC:

$\overrightarrow{OD} = \overrightarrow{OC} + \overrightarrow{CD}$

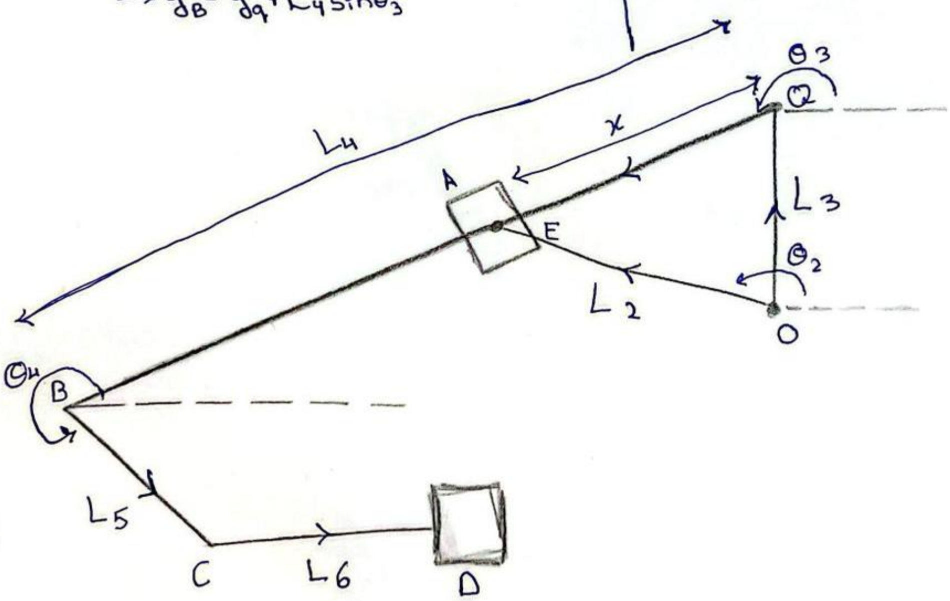
$x_D + i y_D = x_C + i y_c + L_6 e^{i\pi 1}$

equating imaginary parts:

$\rightarrow y_D = y_c +$ [redacted]

equating real parts:

$\longrightarrow x_D = x_c + L_6$



Velocity Analysis:

Point A:

$$V_A = L_2 \omega_2 (ie^{i\theta_2}) \quad , \quad \begin{array}{l} V_A^x = L_2 \omega_2 \sin\theta_2 \\ V_A^y = L_2 \omega_2 \cos\theta_2 \end{array}$$

Point E:

$$\vec{V_E} = \vec{V_A} + \vec{V_{EA}} = x\,\omega_3 (ie^{i\theta_3}) = V_A^x + iV_A^y + V_{EA}(ie^{i\theta_3})\ e^{-i\theta_3}$$

$$-iP\omega_3 = [V_A^x + iV_A^y]\,e^{-i\theta_3} + V_{EA}$$

$$V_{EA} = -V_A^x \cos\theta_3 - V_A^y \sin\theta_3$$

$$\omega_3 = \frac{V_A^x \sin\theta_3 - iV_A^y \cos\theta_3}{x}$$

Point B:

$$\vec{V_B} = \underbrace{\cancel{\vec{V_Q}}}_{0} + \vec{V_{QB}} \quad , \quad V_B = L_4 \omega_3 (ie^{i\theta_3})$$

$$V_B^x = -L_4 \omega_3 \sin\theta_3 \quad , \quad V_B^y = L_4 \omega_3 \cos\theta_3$$

Point C:

$$\vec{V_c} = \vec{V_B} + \vec{V_{CB}} \quad , \quad V_C = V_B^x + iV_B^y + (\omega_4 L_5)\,ie^{i\theta_4}$$

Real: $V_c = V_B^x - \omega_4 L_5 \sin\theta_4$

Img: $\omega_4 = \dfrac{-V_B^y}{L_5 \cos\theta_4}$

Acceleration Analysis:

Point A:

$\overrightarrow{A_A} = (-\omega_2^2 L_2 + i\alpha_2 r_2)e^{i\theta_2} = A_A^x + A_A^y$

$A_A^x = -L_2\omega_2^2 \cos\theta_2 - L_2\alpha_2 \sin\theta_2$

$A_A^y = -L_2\omega_2^2 \sin\theta_2 + L_2 \alpha_2 \cos\theta_2$

Point E:

$\overrightarrow{u_t} = e^{i\theta_3}$, $\overrightarrow{u_n} = -ie^{i\theta_3}$, $A_{EA}^n = -2V_{EA}\omega_3$

$\overrightarrow{A_E} = \overrightarrow{A_A} + \overrightarrow{A_{EA}}$

$\overrightarrow{A_{EA}} = A_{EA}^n \overrightarrow{u_n} + A_{EA}^{SL}\overrightarrow{u_t}$

$\overrightarrow{A_E} = A_A^x + iA_A^y + \overrightarrow{A_{EA}}$

$-\omega_3^2 x = A_A^x \cos\theta_3 + A_A^y \sin\theta_3 + A_{EA}^{SL}$

$$\alpha_3 = \frac{-A_A^x \sin\theta_3 + A_A^y \cos\theta_3 + 2V_{EA}\omega_3}{x}$$

Point B:

$\overrightarrow{A_B} = \overrightarrow{A_Q} + \overrightarrow{A_{QB}}$, $\overrightarrow{A_B} = (-\omega_3^2 L_4 + i\alpha_3 L_4)e^{i\theta_3}$

$A_B^x = -\omega_3^2 L_4 \cos\theta_3 - \alpha_3 L_4 \sin\theta_3$

$A_B^y = -\omega_3^2 L_4 \sin\theta_3 + \alpha_3 L_4 \cos\theta_3$

Point C:

$A_C = \overrightarrow{A_B} + \overrightarrow{A_{CB}}$, $A_C = A_B^x + iA_B^y + \overrightarrow{A_{CB}}$

$A_C = A_B^x + iA_B^y + (-\omega_4 L_5 + i\alpha_4 L_5)e^{i\theta_4}$

Real: $A_C = A_B^x - \omega_4 L_5 \cos\theta_4 - \alpha_4 L_5 \sin\theta_4$

Img: $$\alpha_4 = \frac{-A_B^y + \omega_4 L_5 i \sin\theta_4}{L_5 \cos\theta_4}$$

Force analysis

$I_g = \frac{mL^2}{12}$

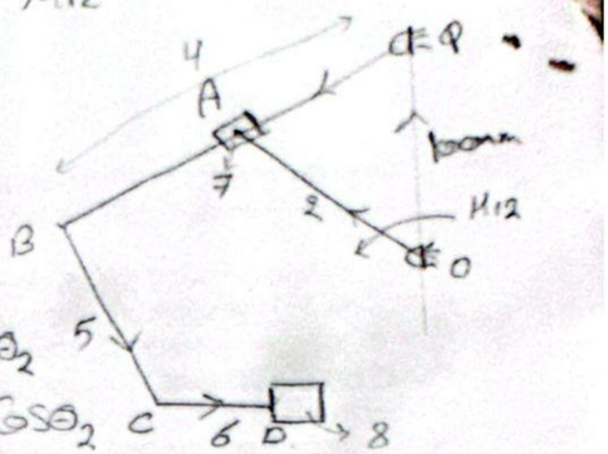
$g_2 = \frac{r_2}{2}$

$g_4 = r_4/2$

$g_5 = r_5/2$

$g_6 = r_6/2$

$M_{12} \rightarrow$ known



$Ag_2 = A_o + A_{g_o}$

$Ag_2 = [-\omega_2^2 g_2 + i\alpha_2 g_2]e^{i\theta_2}$

$Ag_2^x = -\omega_2^2 g_2 \cos\theta_2 - \alpha_2 g_2 \sin\theta_2$

$Ag_2^y = -\omega_2^2 g_2 \sin\theta_2 + \alpha_2 g_2 \cos\theta_2$

$Ag_4 = A_Q + A_{gQ} = [-\omega_3^2 g_4 + i\alpha_3 g_4]e^{i\theta_3}$

$Ag_4^x = -\omega_3^2 g_4 \cos\theta_3 - \alpha_3 g_4 \sin\theta_3$

$Ag_4^y = -\omega_3^2 g_4 \sin\theta_3 + \alpha_3 g_4 \cos\theta_3$

$Ag_5 = AB^x + AB^y J + [-\omega_4^2 g_5 + i\alpha_4 g_5]e^{i\theta_4}$

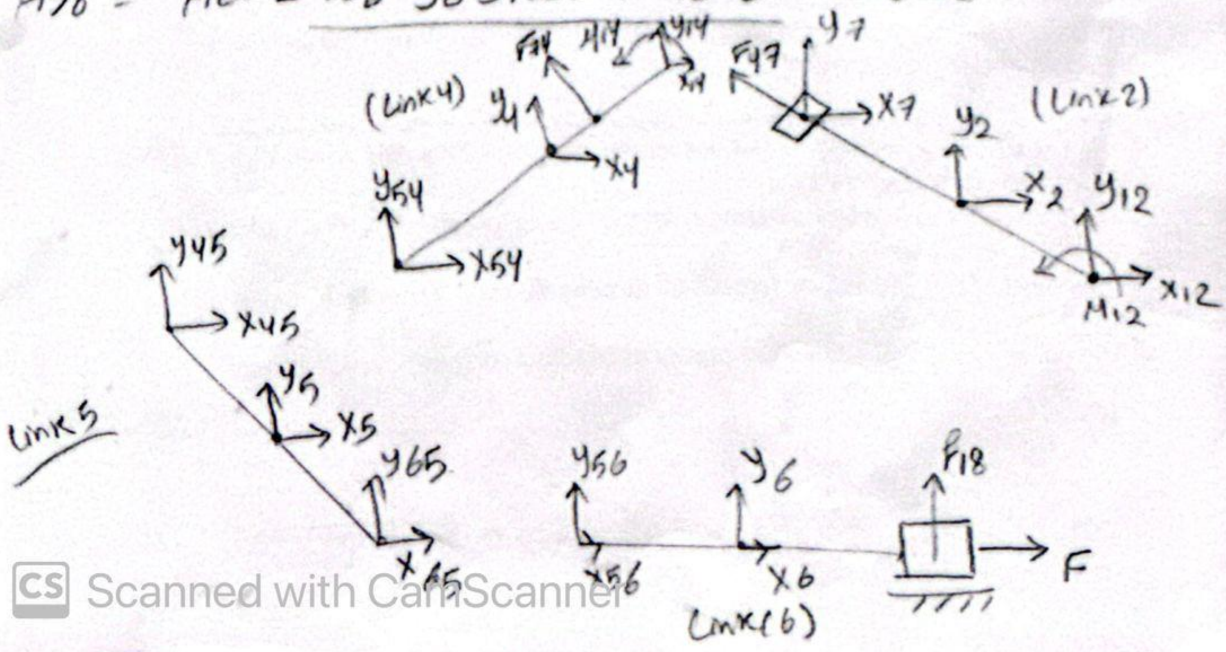
$Ag_5^x = AB^x - \omega_4^2 g_5 \cos\theta_4 - \alpha_4 g_5 \sin\theta_4$

$Ag_5^y = AB^y - \omega_4^2 g_5 \sin\theta_4 + \alpha_4 g_5 \cos\theta_4$

$Ag_6 = Ac^x + Ac^y J + [-\omega_6^2 g_6 + i\alpha_6 g_6]e^{i\theta_6}$

$Ag_6^x = Ac^x - \omega_6^2 g_6 \cos\theta_6 - \alpha_6 g_6 \sin\theta_6$

$Ag_6^y = Ac^y - \omega_6^2 g_6 \sin\theta_6 + \alpha_6 g_6 \cos\theta_6$



Link 2

$\phi = 180 - \theta_2$

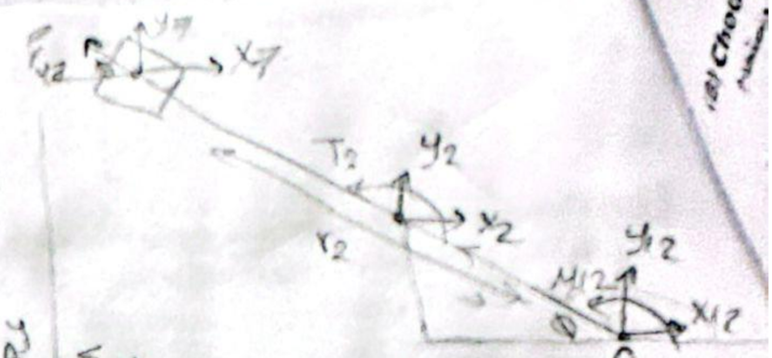
$X_2 = -m_2 A_{g2}^x$

$y_2 = -m_2 A_{g2}^y$

$T_2 = -I_2 \alpha_2$

$X_7 = -m_7 A_A^x$

$y_7 = -m_7 A_A^y$



$\sum M_O = 0 = M_{12} - X_2 \sin\phi\, r_2/2 - y_2 \cos\phi\, r_2/2 + T_2 - (X_7 - F_{47}\cos\phi)\sin\phi\, r_2 - (y_7 + F_{47}\sin\phi)\cos\phi\, r_2 = 0$

$F_{47} = ✓$

$\sum F_y = 0 \to y_{12}$ ✓

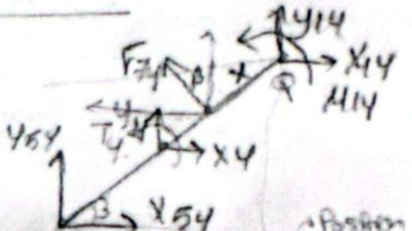
$\sum F_x = 0 \to X_{12}$ ✓

Link 4 $\quad \theta_3 = 180 + \beta$

$\beta = \theta_3 - 180$

$y_4 = -m_4 A_{g4}^y \,,\; X_4 = -m_4 A_{g4}^x$

$F_{74} = -F_{47} \,,\; T_4 = -I_4 \alpha_4$



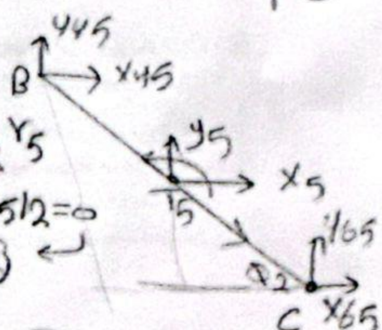
$\sum M_\phi = M_{14} + T_4 - y_{54}\cos\beta\, r_4 + X_{54}\sin\beta\, r_4 - F_{74} * X$ (Position analysis)

$- y_4 \cos\beta\, r_4/2 + X_4 \sin\beta\, r_4/2 = 0 \to (1) < \begin{matrix} y_{54} \\ X_{54} \end{matrix}$

$\sum F_y = 0 \to y_{14} + y_4 + y_{54} + F_{74}\cos\beta = 0 \to y_{14}$ ✓

$\sum F_x = 0 \to X_{14} + X_4 + X_{54} + (-F_{74}\sin\beta) = 0 \to X_{14}$ ✓

Link 5 $\quad \phi_2 = 360 - \theta_4$



$\sum M_C = 0 = -y_{45} * \cos\phi_2\, r_5 - X_{45}\sin\phi_2\, r_5 + T_5 - X_5 \sin\phi_2\, r_5/2 - y_5 * \cos\phi_2\, r_5/2 = 0$

(2)

sub (1) & (2) $\to y_{45} ,\, X_{45}$

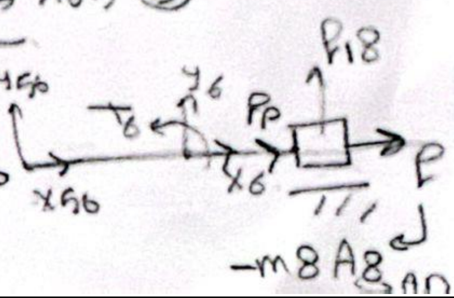
$\sum F_y = 0 = y_{45} + y_{65} + y_5 = 0 \to y_{65}$ ✓

$\sum F_x = 0 = X_{45} + X_5 + X_{65} = 0 \to X_{65}$ ✓

Link 6 $\quad \sum F_y = y_{56} + y_6 + F_{18} = 0$

F_{18} ✓

$\sum F_x = 0 = -F_P - m_8 A_8 + X_6 + X_{56} = 0$



resisting load $\leftarrow$ $F_P = ✓$

I/P. Power = $M_{12} * \omega_2$ watt.

Shaking forces

$$X_{sh} = \sum X_{i1} = -(X_{12} + X_{14} + F_P)$$

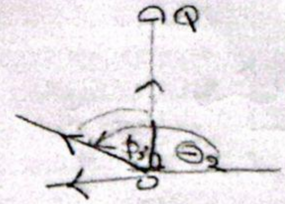
$$Y_{sh} = \sum Y_{i1} = -(Y_{12} + Y_{14})$$

$$M_{sh} = -(M_{12} + M_{14})$$

$$\phi_3 = \theta_2 - 90$$

$$F_{OA} = \frac{M_{12}}{\overline{OA}}$$

$$M_{14} = F_{OA} * \sin \phi_3 * \overline{OQ}$$



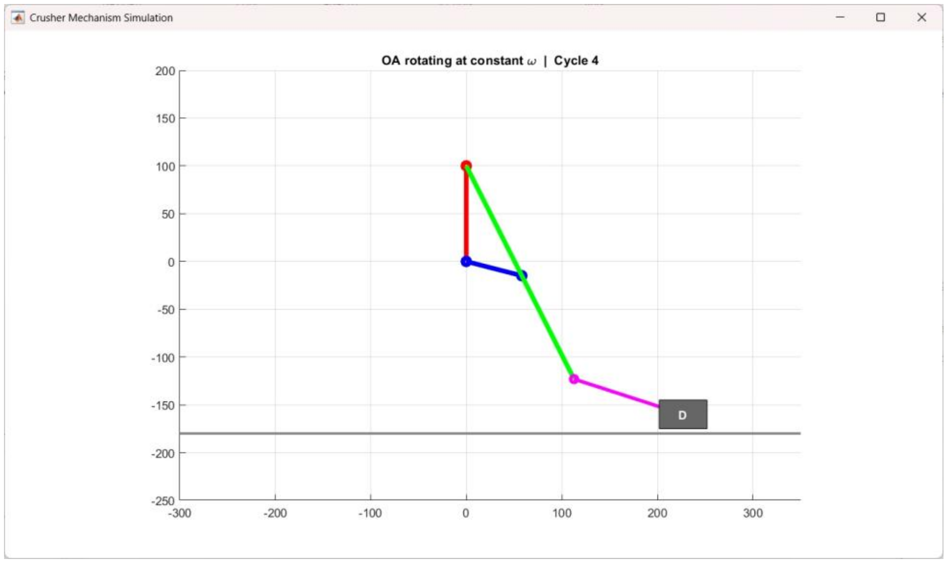
vi. MATLAB simulation:-

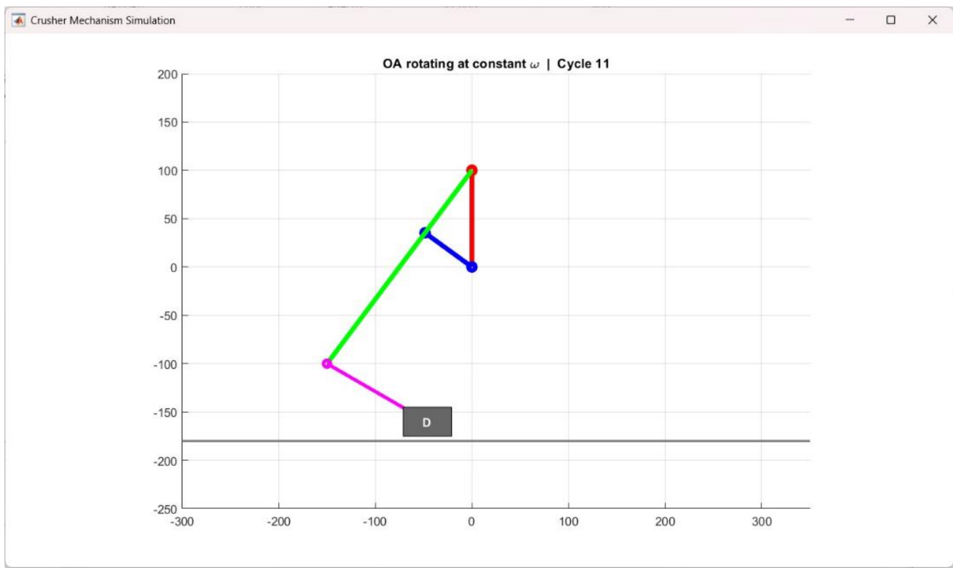
```
>> clear; clc; close all;

% ============================================================

% 1. Mechanism Dimensions (mm)

% ============================================================

OQ = 100;     % Fixed vertical link

OA = 60;      % Driving crank length

QB_len = 250;  % Connecting link

BC = 120;     % Slider connecting rod

CD = 40;      % Crusher block length (visual only)

% ============================================================

% 2. Figure Setup (Clean Window)

% ============================================================

fig = figure( ...

   'Units','normalized', ...

   'Position',[0.1 0.1 0.7 0.7], ...

   'Color','w', ...

   'MenuBar','none', ...

   'ToolBar','none', ...

   'NumberTitle','off', ...

   'Name','Crusher Mechanism Simulation');

hold on; grid on;

% ============================================================

% 3. Constant Angular Speed Definition

% ============================================================
```

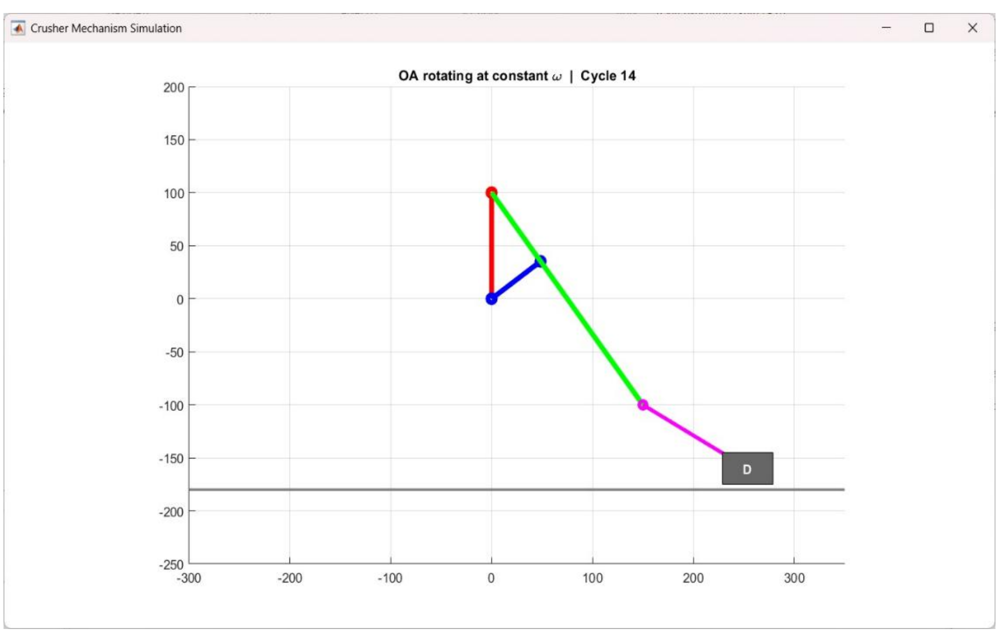
```
omega = 2*pi;    % rad/s (1 revolution per second)
dt = 0.01;      % Time step [s]
theta = 0;      % Initial angle
cycle = 1;      % Cycle counter
% ============================================================
% 4. Continuous Motion Loop
% ============================================================
while ishandle(fig)
  cla;   % Clear previous frame
  % ------------------------------
  % Fixed points
  % ------------------------------
  O = [0 0];
  Q = [0 OQ];
  % ------------------------------
  % OA crank (constant speed)
  % ------------------------------
  A = [OA*cos(theta), OA*sin(theta)];
  % ------------------------------
  % Link QB
  % ------------------------------
  angle_QA = atan2(A(2)-Q(2), A(1)-Q(1));
  B = Q + QB_len * [cos(angle_QA) sin(angle_QA)];
  % ------------------------------
  % Slider constraint (BC)
  % ------------------------------
```

```
y_floor = -160;              % Horizontal slider path
dy = abs(B(2) - y_floor);

if dy <= BC
   dx = sqrt(BC^2 - dy^2);
   C = [B(1) + dx, y_floor];
else
   C = [B(1), y_floor];
end

% ------------------------------
% Crusher attachment (rigid joint)
% ------------------------------
D = C;   % Crusher is rigidly attached to end of BC

% =======================================================
% 5. Drawing
% =======================================================

% Ground line
line([-400 400], [y_floor-20 y_floor-20], ...
   'Color',[0.5 0.5 0.5], 'LineWidth',2);

% Links
plot([O(1) Q(1)], [O(2) Q(2)], 'r-o', 'LineWidth',4); % OQ
plot([O(1) A(1)], [O(2) A(2)], 'b-o', 'LineWidth',4); % OA
```

```
    plot([Q(1) B(1)], [Q(2) B(2)], 'g',   'LineWidth',4); % QB
    plot([B(1) C(1)], [B(2) C(2)], 'm-o', 'LineWidth',3); % BC
    % Pin joint (visual)
    plot(C(1), C(2), 'ko', 'MarkerFaceColor','k', 'MarkerSize',6);
    % Crusher block (centered at C)
    rectangle('Position',[D(1)-25 D(2)-15 50 30], ...
      'FaceColor',[0.4 0.4 0.4]);
    text(D(1)-5, D(2), 'D', 'Color','w', 'FontWeight','bold');
    % ======================================================
    % 6. View settings
    % ======================================================
    axis equal;
    axis([-300 350 -250 200]);
    title(['OA rotating at constant \omega  |  Cycle ', num2str(cycle)]);
    drawnow;
    % ======================================================
    % 7. Time update
    % ======================================================
    theta = theta + omega * dt;
    if theta >= 2*pi
      theta = theta - 2*pi;
      cycle = cycle + 1;
    end
    pause(dt);
  end
```


Crusher Mechanism Simulation
OA rotating at constant ω | Cycle 4
D


Crusher Mechanism Simulation
OA rotating at constant ω | Cycle 11
D


Crusher Mechanism Simulation
OA rotating at constant ω | Cycle 14
D